MIL-I-11525G
13 November 1969
SUPERSEDING
MIL-I-0011525F(MU)
15 February 1968
MIL-I-11525C
16 January 1958

MILITARY SPECIFICATION

IGNITION CYLINDER, FLAME THROWER, PORTABLE, M1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of ignition cylinder.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366	-	Sieve, Test.
TT-W-572	-	Wood Preservative.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-10830	-	Potassium Picrate.
MIL-B-13900	-	Body, Ignition Cylinder.

FSC 1375

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL81-1-454 - Ignition Cylinder, Flame Thrower, Portable, M1.
IDL81-1-454 - Ignition Cyclinder, Flame Thrower, Portable, M1- Inspection Data List.
DL136-13-1110 - Tester Force 0 - 100 Lb.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL81-1-454 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the ignition cylinder shall conform to the specifications and drawings listed on Drawing DL81-1-454 and subsidiary drawings.

3.2 Assembly. The cylinder shall be assembled as specified on Drawing D81-1-454. (See 6.6.)

3.3 Incendiary mixture.

3.3.1 Mixture. The mixture shall be as specified on Drawing B143-13-5 (see 6.10).

3.3.2 Moisture content. The moisture content of the mixture as it is introduced into the cavity shall not exceed 0.15 percent by weight when tested as specified in 4.4.4.1.

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3.4 Starter mixture. The starter mixture shall be as specified on Drawing B143-7-2 (see 6.3).

3.5 Incendiary and starter mixture consolidation. Pressure shall be exerted to consolidate the mixtures in the cavity as shown on Drawing D81-1-454 (see 6.4). Upon functioning, the consolidated incendiary mixture shall remain in the cylinder until completely consumed.

3.6 Match assembly.

3.6.1 Match. The striker mixture of the match (Drawing C81-1-9418) shall be applied in such a manner to insure functioning of the cylinder when the match is actuated (see 6.5).

3.6.2 Match assembly. The force required to dislodge the ring from the match pin groove shall be between 20 and 75 pounds when tested as specified in 4.4.4.2.

3.7 Lifting charge. A closure plate lifting charge of potassium picrate, conforming to MIL-P-10830 shall be placed in each cavity as shown on Drawing D81-1-454. Particle size shall conform to table I (see 6.11).

Table I. Particle size

US standard sieves (RR-S-366)	Percent retained
10	0.5 max
18	45.0 min
60	40.0 max
pan	15.0 max

3.8 Sealing.

3.8.1 Match assembly. The specified water-proofing sealant shall be applied to the match pin in such a manner that it will fill the match hole counter-bore and cover the restraining ring to form a positive seal against moisture (see 6.9(b)).

3.8.2 Closure plate assembly. The closure plate assembly bonding shall be accomplished in a manner to assure waterproofing (see 6.9(a)).

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3.9 Functioning. The cylinder shall function and emit a jet flame from each cavity for 8 to 12 seconds when tested as follows:

(a) Without having been immersed in accordance with 4.4.4.3(a).

(b) After immersion in accordance with 4.4.4.3(b).

Burning of residuals or nonincendiary components shall not be considered in reckoning the burning time.

3.9.1 High temperature. The cylinder shall function in accordance with 3.9(a) after having been subjected to a temperature of 160° + 5° F for 72 + 2 hours and then having been subjected to a temperature of 70° ± 20° F for a minimum period of 12 hours.

3.9.2 Low temperature. The cylinder shall function in accordance with 3.9.(a) after having been subjected to a temperature of minus 65° + 5° F for 72 + 2 hours and then having been subjected to a temperature of 70° ± 20° F for a minimum period of 12 hours.

3.10 Container (Drawing C81-2-13). The filled container shall not leak when tested as specified in 4.4.4.4(see 6.8).

3.11 Preproduction. Prior to the start of regular production, a preproduction lot of ignition cylinders shall be produced in accordance with this specification for examination and tests (see 4.3).

3.12 Workmanship. The cylinder shall be free from contamination (foreign matter, oil or grease) and damage such as cracks, or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lot.

(a) Cylinders and containers. A preproduction lot of 1500 cylinders, packaged as specified in 5.1, shall be manufactured using the same methods, materials, equipment, processes and procedures as will be used during regular production.

(b) Match assembly. A preproduction lot of 8000 match assemblies (less striker mix) shall be manufactured using the same methods, materials, equipment, processes and procedures as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests.

(a) Containers. Seventy-five containers of ignition cylinders shall be randomly selected from the preproduction lot.

(b) Incendiary mixture. A 10-gram sample shall be taken at random from each batch (see 6.7) of incendiary mixture used to produce the preproduction lot.

4.3.2.2 For destructive tests.

(a) Cylinders. One hundred and fifty cylinders shall be taken from the 75 containers (4.3.2.1(a)).

(b) Match assembly. Two hundred match assemblies shall be randomly selected from the preproduction lot.

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4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests.

(a) Containers. The sample containers and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects. The AQL's do not apply.

(b) Incendiary mixture. The sample of mixture shall be tested in accordance with 4.4.4.1.

4.3.3.2 For destructive tests.

(a) Cylinders. The sample cylinders shall be tested in accordance with table II.

(b) Match assembly. The sample match assemblies shall be tested in accordance with table II.

Table II. Preproduction sampling, inspection, acceptance/rejection criteria

Quantity	Requirement	Test	Attributes	Cavities	
				Accept	Reject
125 cavities (25 cylinders)	3.9.1	4.4.4.3(a)	*Duds, burning time failure or simultaneous ignition	6	7
125 cavities (25 cylinders)	3.9.1	4.4.4.3(b)		6	7
125 cavities (25 cylinders)	3.9.2	4.4.4.3(a)	*Duds, burning time failure or simultaneous ignition	6	7
125 cavities (25 cylinders)	3.9.2	4.4.4.3(b)		6	7
125 cavities (25 cylinders)	3.9.3	4.4.4.3(a)	*Duds, burning time failure or simultaneous ignition	6	7
125 cavities (25 cylinders)	3.9.3	4.4.4.3(b)		6	7
200 Match assemblies	3.6.2	4.4.4.2	Force required to dislodge ring incorrect	**20	21

*The preproduction sample shall be rejected, if two consecutive cavities are duds or three or more defects are found in any one cylinder (see 6.12).
**The preproduction sample shall be rejected if 11 or more defectives are less that the lower limit or 11 or more defectives are above the high limit.

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4.3.4 Acceptance/rejection criteria. The preproduction samples shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The acceptance number is zero. The suppliers shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the cylinders produced by one manufacturer, in no more than five consecutive work days by the personnel of one 8-hour shift only and identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of cylinders, no more than one interfix number from one manufacturer of the following components is permitted.

(a) Incendiary mixture
(b) Starter mixture

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests.

(a) Incendiary mixture. A 10-gram sample shall be taken at random from each batch (see 6.7) of mixture.

(b) Classification of defects. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests.

(a) Match assembly. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-4.

(b) Functioning. Sampling shall be conducted in accordance with MIL-STD-105, general inspection level I. The unit of product is one cavity.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests.

(a) Incendiary mixture. The sample of mixture shall be tested in accordance with 4.4.4.1. The acceptance number is zero.

(b) Classification of defects. The sample cylinders and the level A preparation for delivery shall be inspected in accordance with the classification of defects and MIL-STD-105.

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4.4.3.2 For destructive tests.

(a) Match assembly. Sample match assemblies shall be tested in accordance with 4.4.4.2 and MIL-STD-105, using an AQL of 4.0 percent defective, applied at each end of the range specified in 3.6.2.

(b) Functioning. Sample cylinders shall be tested in accordance with 4.4.4.3(b), MIL-STD-105, and table III.

Table III. Acceptance/rejection criteria

*Attribute	AQL
Failure of cavity to burn	2.5 percent defective
Burning time failure	2.5 percent defective
Simultaneous ignition of two cavities	2.5 percent defective

*The lot shall be rejected if two consecutive cavities are duds or three or more defects are found in any one cylinder.

4.4.3.3 Classification of defects. (See inspection data list IDL81-1-454 for inspection equipment required to perform the examinations specified in this section.)

(a) Cylinder, ignition, portable, flame thrower, M1 (Drawing D81-1-454).

Categories		Defects
Critical:		None defined
Major:	AQL 4.0 percent defective	
101		Component missing or incorrectly assembled
102		Location of sealant incorrect
103		Pin extended beyond rim of cylinder
104		Pin not coated as specified
105		Cylinder not waterproofed as specified
106		Workmanship (3.12)

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4.4.3.3 Classification of defects (continued).

(b) Container, ignition cylinder packing (Drawing C81-1-13).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL 0.25 percent defective		
101		Leakage	4.4.4.4
	AQL 1.0 percent defective		
102		Contents missing	
103		Marking illegible or incorrect	
104		Key missing	
105		Coating inadequate or incorrect	
106		Workmanship (3.12)	

(c) Preparation for delivery (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Loose pack
102		Container incorrect
103		Marking missing or illegible
104		Strapping missing or loose

4.4.4 Tests. Tests shall be conducted as follows: (See inspection data list IDL81-1-454 for inspection equipment required to perform the tests specified in this section.)

4.4.4.1 Mositure content. The moisture content shall be determined by weighing to the nearest 0.1 milligram, the sample of mixture in a covered Petri dish. Spread the sample evenly in a thin layer. Place the Petri dish with the sample in a vaccum desiccator, remove dish cover, and allow the sample to remain in the desiccator for 24 hours at ambient temperature and less than 380 millimeters of mercury. The desiccator shall contain a desiccant such as Drierite or silica gel which shall be always fresh (dry) or new material. The total loss in weight of the sample shall be considered as moisture in the calculation of moisture content. When sample is removed from desiccator to be reweighed, place cover on dish to prevent sample from picking up moisture.

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4.4.4.2 Match assembly. The match pin assembly shall be placed in the test fixture, Drawing D136-13-1110, and the force required to strip the ring from the pin shall be recorded.

4.4.4.3 Functioning.

(a) Normal atmosphere. The ignition cylinder shall be brought to atmospheric temperature and then functioned as specified in 3.9.1 using a modified M7 gun mechanism (see figure 1) at that temperature. Cavities shall not be fired to burn simultaneously (see 6.13).

(b) Water immersion. The ignition cylinder shall be brought to atmospheric temperature and be immersed in clear water at a temperature of 75° ± 5°F to a covering depth of 12 to 14 inches for a period of 24 hours. On removal from the water, the cylinders shall be functioned as specified in 3.9.1 using a modified M7 type gun mechanism (see figure 1) at atmospheric temperature. Withdrawal from the water shall be regulated to permit functioning of each cylinder within 1 hour after removal from the water.

4.4.4.4 Container. The container and contents when at room temperature shall be submerged in water at 185° ± 15°F to a depth of 2 to 3 inches below the surface for a minimum period of 20 seconds. Evidence of bubbles shall indicate leakage (see 6.8).

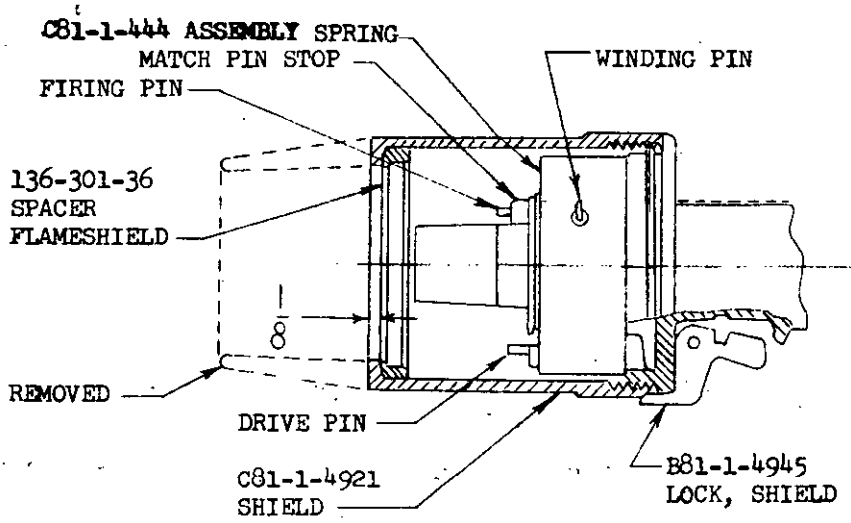
5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. The cylinders shall be packaged as specified on Drawing C81-2-13.

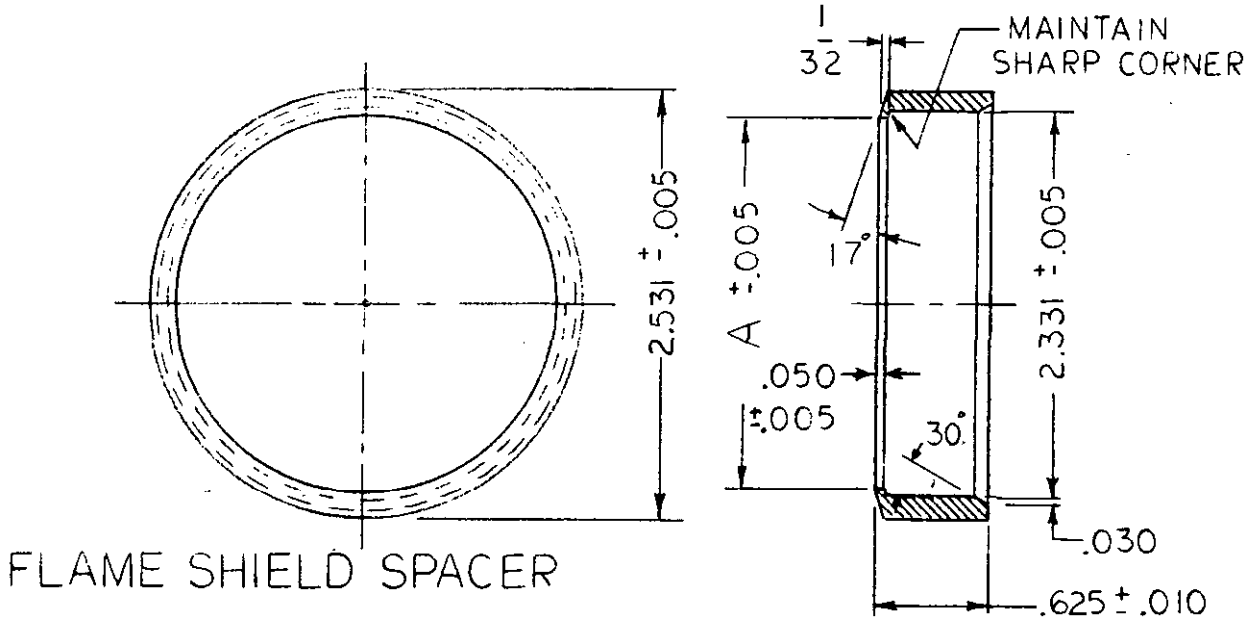
5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. Fifty cans of ignition cylinders (two per can) packaged as specified in 5.1 shall be packed in two layers in an overseas type or class container conforming to PPP-B-621, except box manufacturer's identification marking, section 3, is not required. The inside dimensions of the container shall be approximately 12-7/8 by 12-7/8 by 8-5/16 inches. The boxes or wood parts shall be preserved by completely immersing in wood preservative conforming to composition A of TT-W-572 for a minimum period of 3 minutes. Prior to packing the ignition cylinders in the treated boxes, the preservative shall be dry as evidenced by the dryness test in MIL-B-2427. Fiberboard pads fabricated from material conforming to type CF, class-weather resistant, variety SW, grade W5c or W6c of PPP-F-320 shall be used between each layer, and the contents and the sides, ends, top and bottom of the box, using additional pads as required to make a tight pack. Each of

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TEST GUN DETAILS



FLAME SHIELD SPACER

NOTE: TEST GUN WILL COMPRISE IGNITION CYLINDER FIRING COMPONENTS OF GUN, PORTABLE FLAME THROWER, M7, D81-1-4920, WITH MODIFIED FLAME SHIELD (TRUNCATED TO EXPOSE JET-FLAME TO VIEW) BUT WILL HAVE NO FLUID PRESSURE MECHANISM DETAILS OTHER THAN THE PISTOL GRIP.

FIGURE 1

GUN, PORTABLE FLAME THROWER, M7, MODIFIED

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two keys conforming to Drawing D22-129-1 shall be placed in a vertical position, point down, perforating if necessary to the pad separating the can layers, between the cans and the side filler pads on opposite sides of every container. Closure of the box shall be in accordance with the requirements of the applicable container specification and its appendix, using galvanized strapping.

5.2.2 Level B. Fifty cans of ignition cylinders packages as specified in 5.1 shall be packed as specified in 5.2.1 except that domestic type or class containers and coated-finish strapping shall be used.

5.3 Palletization. Unless otherwise specified in the contract or order, cylinders packed in accordance with 5.2.1 shall be palletized in accordance with Drawing D81-1-9721 (see 6.2).

5.4 Marking. In addition to any special marking required by the contract or order, marking shall be as follows (see 5.2.1, dryness test):

(a) Each unit package shall be marked in accordance with Drawing C81-2-13.

(b) Each shipping container (wood box) shall be marked in accordance with Drawing D81-1-9652.

(c) Palletized loads shall be marked in accordance with Drawing D81-1-9721.

6. NOTES

6.1 Intended use. The ignition cylinder is used in the portable flame thrower gun to ignite the thickened fuel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of the specification.
(b) Shipment group size and marking (when required).
(c) M7 gun required for test will be Government furnished.
(d) Level of packing required and whether item should be palletized.
(e) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

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6.3 Starter mixture. The following has been found to be a satisfactory proceedure. Granulate the mixture by passing it through a No. 30 US standard sieve conforming to RR-S-366. Allow the mixture to dry until stiff but still moist to the touch. It has been found that 0.53 + 0.1 gram is usually sufficient for the total weight of starter mixture for one cavity. This information is advisory and it is understood that the requirements of 3.9.1 must be met regardless of the weights of starter mixture used.

6.4 Consolidation. A minimum load of 225 pounds per cavity transmitted through a flat bottom ram not less than 27/64 inch in diameter has been found to be satisfactory.

6.5 Scratch mixture. The following scratch mixture formulation and process have been found suitable for application:

Scratch mixture

Carburundum	20 pts/100
Red phosphorous	80 pts/100

Resin solvent

Ethyl alcohol (MIL-E-463, grade 5)	99.75 pts/100
Sodium Hydroxide (MIL-S-11795) 20 percent aqueous solution	0.25 pts/100

Liquid binder (resin solution)

Resin solvent	93 pts/100
Phenolparaformaldehyde	7 pts/100

or

Liquid binder solution

Styron 666 (small pellet variety)	40 pts/100
Benzene (VV-B-231)	60 pts/100

Additional benzene may be added for thinning.

Add the scratch mixture to the liquid binder in proportions such that the resulting slurry will be of a viscosity suitable for coating the match pin in accordance with Drawing C81-1-9418. Tempering of the match assembly at 175° C (dry heat) for 15 minutes following the installation of the restraining ring and directly after application of the phosphate

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coating, has been found helpful in controlling the range of forces required to dislodge the restraining ring from the match pin groove.

6.6 Precautions. Due to the hazardous nature of filling cylinders with mixtures and employing volatile solvents, suppliers are advised to take all necessary precautions to protect personnel from injury while filling and assembling cylinders.

6.7 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.8 Container repair. Containers showing evidence of leakage may be reconditioned by wiping dry immediately as withdrawn from the test bath and at peak temperature (to prevent moisture from being drawn into the container), sealing by soldering or other suitable means and retesting.

6.9 Seals.

(a) Closure plate. The seal may be pressed into each of the five cavities or forced into position together with the closure plate. Precautions against tearing of the foil in the effective sealing areas must be observed. An alternative method of applying the seal is as follows: The cylinder subassembly may be positioned in a press on a suitable resilient protective pad. A strip or blank of 0.005 ± 0.001 lead foil which has been precoated with the specified waterproofing scalant or adhesive on its cylinder side may be fed over the exposed end and struck with a suitable die to coin, form, and fixedly conform the lead foil to the closure plate recesses. The exposed foil in the closure plate recesses may then be lightly coated with the specified adhesive sealant and the closure plates carefully placed and pressed into position. Following the setting of the adhesive the remaining exposed foil may be removed in a manner not to disturb the seals and a coating or coatings of the specified waterproofing sealant applied to the closure end of the cylinder. The central bore and sides of the finished cylinder should preferably be free of sealant. Caution must be exercised in applying solvent type sealants and adhesives to preclude the trapping of solvents within the cavities, where they may attack other adhesives or exert undue gaseous pressure.

(b) Match. It has been found advantageous to apply the waterproofing sealant to the match pin adjacent to the restraining ring position at the scratch-mix side during the assembly of the pin into the cylinder. The quantity of sealant should be such that it will fill the match pin hole counterbore to cover the restraining ring upon pressing the match assembly to seat.

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6.10 Weight of incendiary mixture. It has been found that 2.75 ± 0.25 grams is usually sufficient for the total weight of mixture for one cavity. This information is advisory and it is understood that the requirements of 3.9.1 must be met regardless of the weight of mixture used.

6.11 Weight of lifting charge. It has been found that 0.11 ± 0.01 gram of potassium picrate is usually satisfactory for this purpose. This information is advisory and it is understood that sufficient charge must be present to lift or expel the closure plate. As an alternate to controlling the particle size as indicated in table I, procedure may be carried out as follows: Place a drop of binder comprising 92 parts of acetone and 8 parts of cellulose nitrate (MIL-N-244, grade D) on the starter mixture and add the potassium picrate while the binder solution is still wet. The components should be completely dry prior to the lead foil assembly operation.

6.12 Dud. A dud, for the purpose of this specification, is defined as the failure of a cavity to ignite due to mechanical or pyrotechnic malfunction.

6.13 Gun cleaning. Prior to each test of 100 cylinders (or as conditions warrant) the gun may be reconditioned in accordance with the provisions of Technical Manual Lubrication Order LO 3-1040-211-30 and by checking and replacing any parts which may contribute to malfunction. Upon the conclusion of burning of each cylinder it has been found helpful to perform the following:

(a) Remove the cylinder.
(b) Clean the shield nozzle (especially at the inner shoulder).
(c) Clean the spacer and the match pin stop (at the match pin hole).
(d) Check the spring case assembly.
(e) Assemble a new cylinder in gun and check for positive cylinder indexing.

6.14 Incendiary mix moisture content. When the moisture content of a batch of mix surpasses the requirement (3.3.2), it may be returned to the drier and then retested.

6.15 Component specification. The following specification is required to complete the M1 ignition cylinder:

MIL-B-13900 - Body, Ignition Cylinder.

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Custodians:

Army - MU
Navy - OS

Review activity:

Navy - OS

User activity:

Navy - MC

Preparing activity:

Army - MU(EA)

Project No. 1375-0003

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